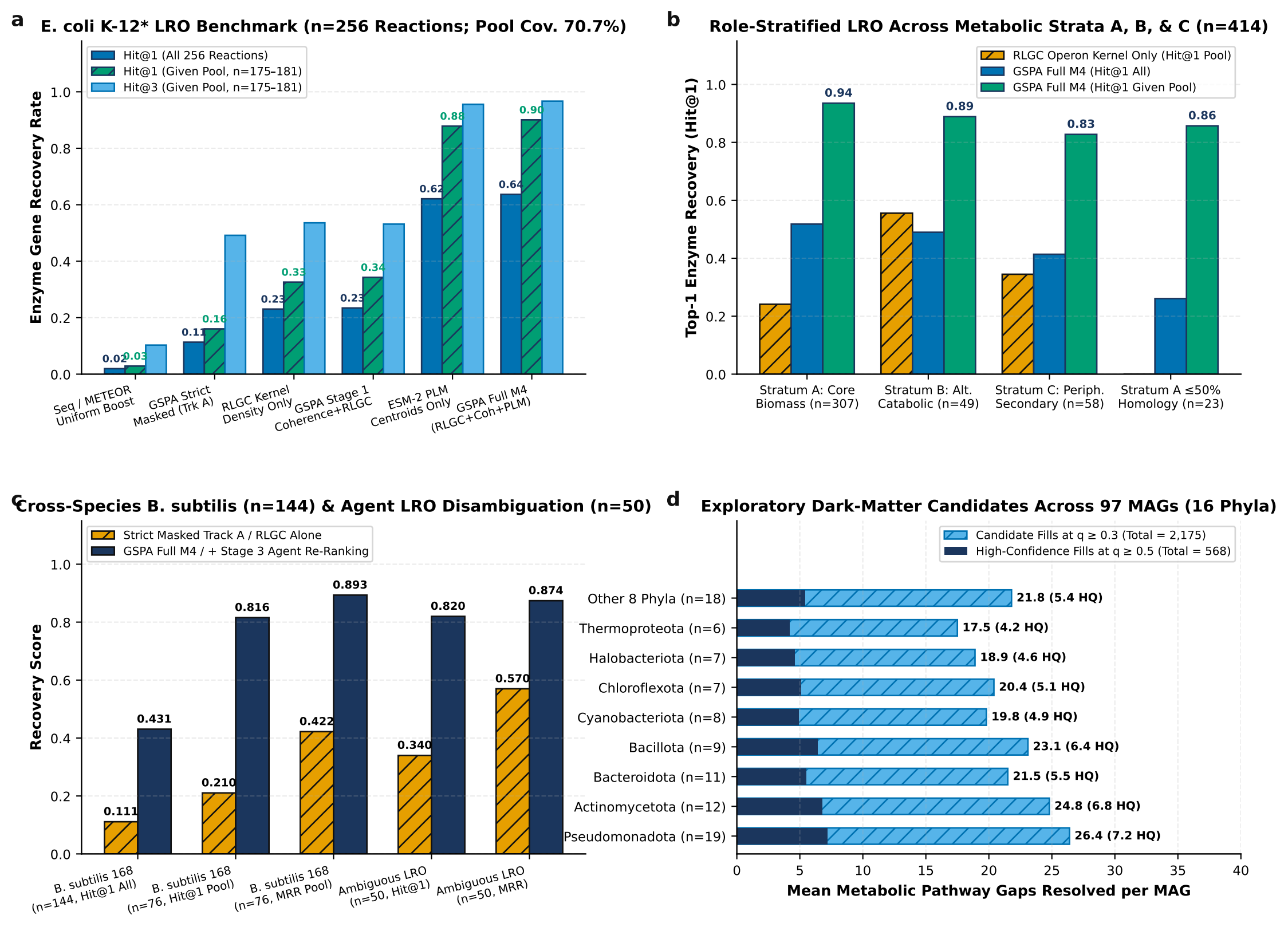
a
E. coli K-12* LRO Benchmark (n=256 Reactions; Pool Cov. 70.7%)
Hit@1 (All 256 Reactions)
Hit@1 (Given Pool, n=175–181)
Hit@3 (Given Pool, n=175–181)
Enzyme Gene Recovery Rate
0.0
0.2
0.4
0.6
0.8
1.0
0.02
0.03
0.11
0.16
0.23
0.33
0.23
0.34
0.62
0.88
0.64
0.90
Seq / METEOR Uniform Boost
GSPA Strict Masked (Trk A)
RLGC Kernel Density Only
GSPA Stage 1 Coherence+RLGC
ESM-2 PLM Centroids Only
GSPA Full M4 (RLGC+Coh+PLM)
b
Role-Stratified LRO Across Metabolic Strata A, B, & C (n=414)
RLGC Operon Kernel Only (Hit@1 Pool)
GSPA Full M4 (Hit@1 All)
GSPA Full M4 (Hit@1 Given Pool)
Top-1 Enzyme Recovery (Hit@1)
0.0
0.2
0.4
0.6
0.8
1.0
0.94
0.89
0.83
0.86
Stratum A: Core Biomass (n=307)
Stratum B: Alt. Catabolic (n=49)
Stratum C: Periph. Secondary (n=58)
Stratum A ≤50% Homology (n=23)
c
Cross-Species B. subtilis (n=144) & Agent LRO Disambiguation (n=50)
Strict Masked Track A / RLGC Alone
GSPA Full M4 / + Stage 3 Agent Re-Ranking
Recovery Score
0.0
0.2
0.4
0.6
0.8
1.0
0.111
0.431
0.210
0.816
0.422
0.893
0.340
0.820
0.570
0.874
B. subtilis 168 (n=144, Hit@1 All)
B. subtilis 168 (n=76, Hit@1 Pool)
B. subtilis 168 (n=76, MRR Pool)
Ambiguous LRO (n=50, Hit@1)
Ambiguous LRO (n=50, MRR)
d
Exploratory Dark-Matter Candidates Across 97 MAGs (16 Phyla)
Candidate Fills at q ≥ 0.3 (Total = 2,175)
High-Confidence Fills at q ≥ 0.5 (Total = 568)
Other 8 Phyla (n=18)
21.8 (5.4 HQ)
Thermoproteota (n=6)
17.5 (4.2 HQ)
Halobacteriota (n=7)
18.9 (4.6 HQ)
Chloroflexota (n=7)
20.4 (5.1 HQ)
Cyanobacteriota (n=8)
19.8 (4.9 HQ)
Bacillota (n=9)
23.1 (6.4 HQ)
Bacteroidota (n=11)
21.5 (5.5 HQ)
Actinomycetota (n=12)
24.8 (6.8 HQ)
Pseudomonadota (n=19)
26.4 (7.2 HQ)
0
5
10
15
20
25
30
35
40
Mean Metabolic Pathway Gaps Resolved per MAG